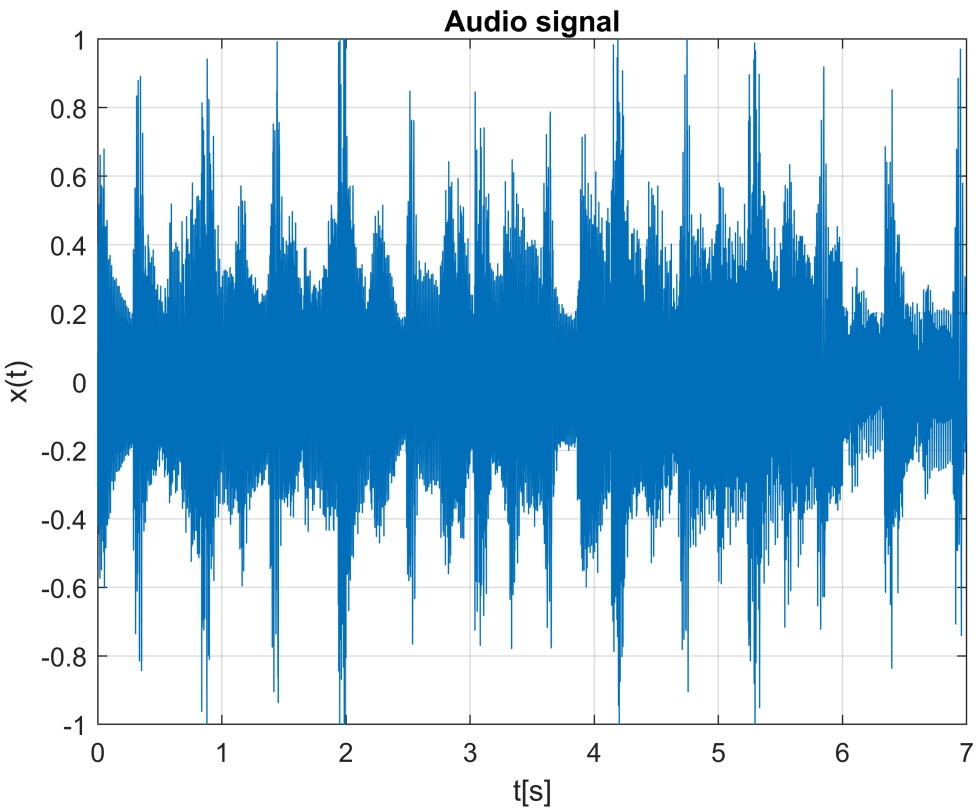
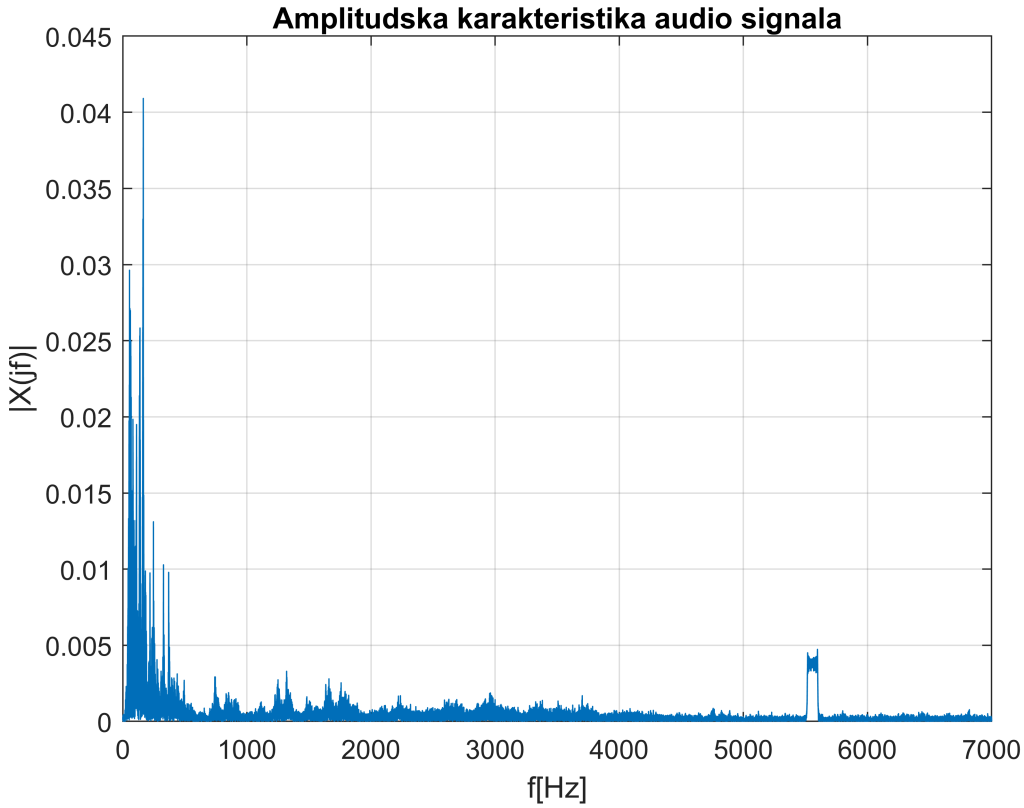
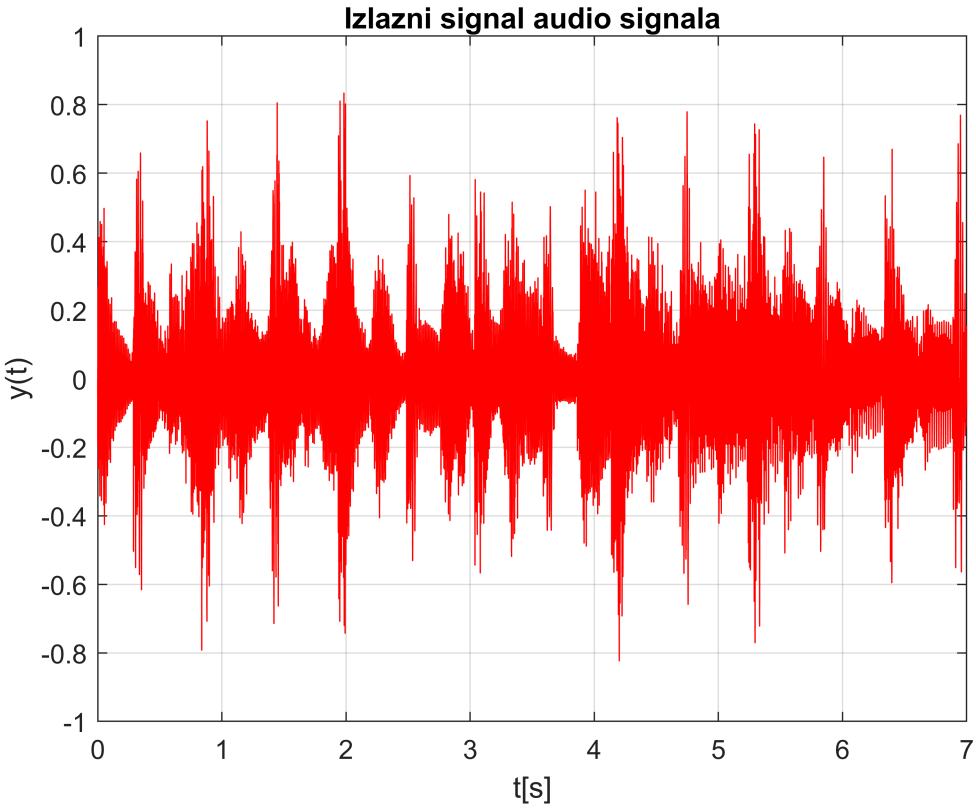
```
clear all; close all; clc;

[x, Fs] = audioread('R2_zasumljen.wav');
t = 0 : 1/Fs : (length(x)-1)/Fs;

figure(1);
plot(t, x);
xlabel('t[s]'); ylabel('x(t)'); title('Audio signal'); grid on;
```



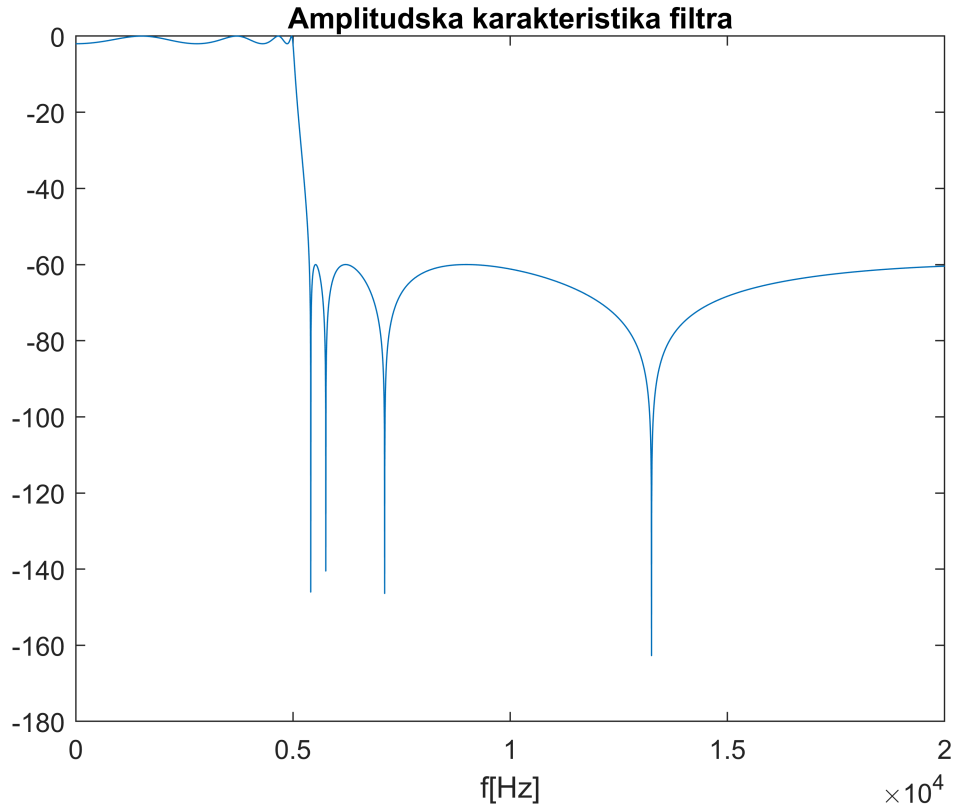

```
%-------------------------------------------------------------
N = 2^nextpow2(length(x));
f1 = 0 : Fs/N : Fs/2;
X = fft(x, N) / length(x);
X1 = abs(X(1 : N/2+1));
X1(2 : N/2+1) = 2*X1(2 : N/2+1);

figure(2);
plot(f1, X1);
xlim([0, 7000]); %da bi se video sum
xlabel('f[Hz]'); ylabel('|X(jf)|'); grid on;
title('Amplitudska karakteristika audio signala');
```



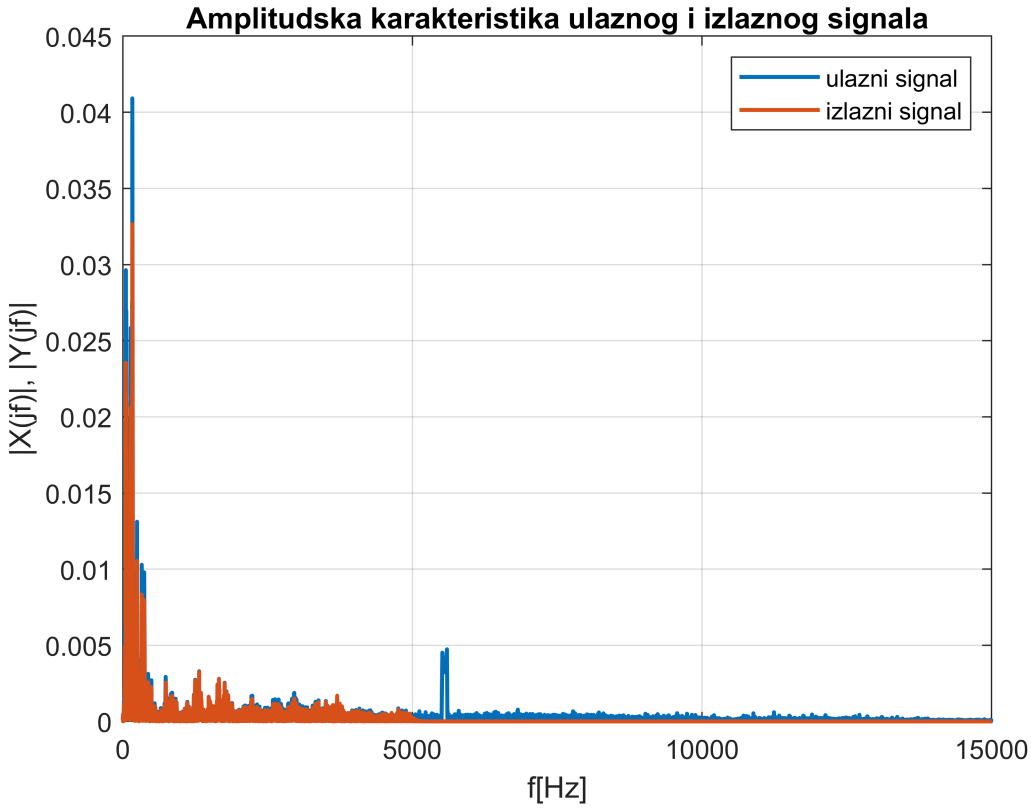

```
%Sinusoidalni sum je negde izmedju 5,5 i 6 KHz
%-------------------------------------------------------------
%Projektujemo digitalni elipticki filtar

Wp = 5000 / (Fs/2);
Ws = 5500 / (Fs/2);
Rp = 2;
Rs = 60;
[n, Wn] = ellipord(Wp, Ws, Rp, Rs);
[b, a] = ellip(n, Rp, Rs, Wp);
[h, f] = freqz(b, a, N/2+1, Fs);

y = filter(b, a, x); %filtriranje signala
Y = fft(y, N) / length(y);
Y1 = abs(Y(1 : N/2+1));
Y1(2 : N/2+1) = 2*Y1(2 : N/2+1);

figure(3);
plot(t, y, 'r');
xlabel('t[s]'); ylabel('y(t)'); grid on;
title('Izlazni signal audio signala');
```



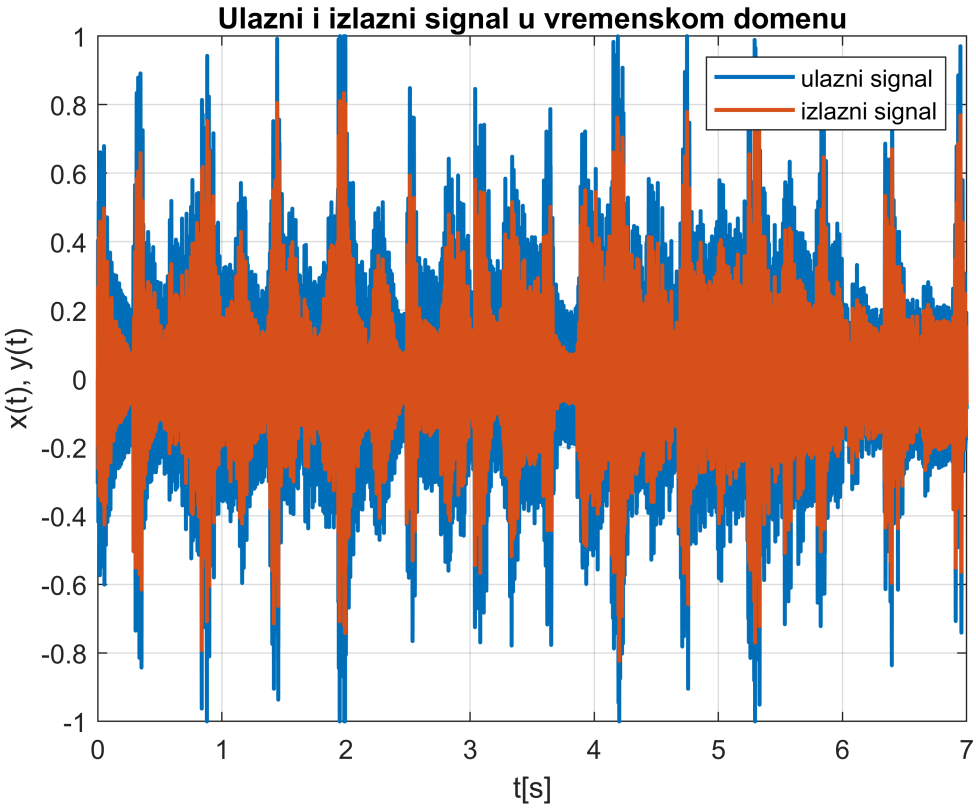

```
%filename = 'C:\Users\38166\Desktop\isfiltriran1.wav';
%audiowrite(filename, y, Fs);
%sound(y, Fs);

%Red filtra nam je 8.
%-------------------------------------------------------------
figure(4);
plot(f, 20*log10(abs(h)));
xlim([0 20000]);
xlabel('f[Hz]'); title('Amplitudska karakteristika filtra');
```




```
figure(5);
plot(f1, X1, 'LineWidth', 1.5); hold on;
plot(f1, Y1, 'LineWidth', 1.5);
xlim([0 15000]);
xlabel('f[Hz]'); ylabel('|X(jf)|, |Y(jf)|');
legend('ulazni signal', 'izlazni signal'); grid on;
title('Amplitudska karakteristika ulaznog i izlaznog signala');
```




```
figure(6);
plot(t, x, 'LineWidth', 1.5); hold on;
plot(t, y, 'LineWidth', 1.5);
xlabel('t[s]'); ylabel('x(t), y(t)');
legend('ulazni signal', 'izlazni signal'); grid on;
title('Ulazni i izlazni signal u vremenskom domenu');
```


Ulazni i izlazni signal u vremenskom domenu
ulazni signal
izlazni signal
x(t), y(t)
t[s]
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
1
2
3
4
5
6
7